

ABOUT

Security and Information Technology engineer. Since 2025 on a ZTNA SaaS platform (TypeScript/Node.js), autonomous and in direct contact with the CTO. Backend/platform: identity federation (Keycloak, Entra ID), RBAC, OpenAPI contracts consumed by the mobile, SDK and C++ network teams. Took the central console from prototype to a self-hostable solution, shipped to production.

PROFESSIONAL EXPERIENCE

Full-Stack Engineer (fixed-term contract) – Snowpack (ZTNA platform)05/2026 - Present

Paris

- **3–6× lower API latency** (avg, p50, p99; k6 load tests): upgraded to Node 24, unblocked by migrating the backend from uncompiled ts-node to compiled ESM (cyclic dependencies resolved).
- **Production incident:** CPU spikes (40–70% every 10 min) traced in Grafana to a gzip bomb; capped decompression size, restricted the API to JSON, reworked the cron jobs. CPU back to 10–20%.
- **Self-hostable console:** shipped to customers as a single executable (esbuild bundle + Node SEA), tested on Docker and Raspberry Pi; wrote the customer docs and setup script.
- **Audit log:** records authentication (success/failure) and tenant operations (create, read, update, delete) from every console endpoint, written asynchronously on the task queue built during the internship.
- **Fixed TOCTOU races** on create/update paths: replaced application read-then-write checks with MongoDB unique indexes.
- **Stack:** TypeScript, Node.js, ESM, swc, esbuild, Node SEA, k6, Grafana, Docker, Vitest, GitLab CI.

INTERNSHIP: Full-Stack Developer – Snowpack (ZTNA platform)02/2025 - 03/2026

Paris

- **Group management, idea to production in 2 months:** from a one-line CTO idea, owned the spec and implementation; users and devices assigned to groups sharing one config, services and licences propagated automatically and recomputed on any membership or policy change.
- **Identity federation** (Keycloak, Microsoft Entra ID): one shared SSO flow and callback for both providers; users and group synced into the console on every SSO login.
- **Typed task queue:** event types checked at compile time, one dispatcher per queue, async and sync modes; reused by another developer without help.
- **Central console** (API + web): single entry point for all Snowpack components; its OpenAPI contracts are consumed by the mobile, SDK and C++ network teams.
- **4× faster load under 2G throttling** (React): removed duplicate fetches across contexts, added caching, narrowed queries.
- **Stack:** TypeScript, Node.js, TSOA / OpenAPI, React, MongoDB, TypeORM, Keycloak, Docker, GitLab CI.

PROJECTS

RFP proposal scorer – 24h hackathon - architecture lead09/2026

Frankfurt

- **Multi-stage LLM scoring pipeline:** regex for non-semantic signals, LLM extraction with verbatim citations, three parallel scoring stages to bound context and cost per call; model lists weaknesses before scoring, so every score cites evidence.
- **Stack:** Python (FastAPI), React, Docker.
- **GitHub:** github.com/UNtergas/SiviHack-2026-Team

EDUCATION

INSA Centre Val de Loire – Engineering degree, Security and Information Technology2020 - 2026

Bourges (2022-2026) · Selective Franco-Vietnamese integrated preparatory programme, Hué (2020-2022)

SKILLS

Languages: TypeScript, JavaScript, Python, SQL
Backend: Node.js, Express, TSOA, REST, OpenAPI / Swagger, FastAPI
Frontend: React, Next.js, Vite
Databases: MongoDB, TypeORM, SQL
Security & Identity: Authentication, RBAC, application security, Keycloak, Microsoft Entra ID
Testing: Vitest (TDD, end-to-end), k6 (load testing)
Build & DevOps: Docker, GitLab CI, ESM, swc, esbuild, Node SEA, Grafana
Practices: OOP, TDD